



**THE UNIVERSITY
OF QUEENSLAND**
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____

Seat Number _____

Student Number

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Family Name _____

First Name _____

School of Mathematics & Physics EXAMINATION

Semester Two Final Examinations, 2021

MATH1061 Discrete Mathematics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Reading Time: 10 minutes

Exam Conditions:

This is a Closed Book examination - no written materials permitted

Casio FX82 series or a calculator on the UQ approved list

During reading time (= planning time) - students are encouraged to review and plan responses to the exam questions

This examination paper will be released to the Library

Materials Permitted In The Exam Venue:

(No electronic aids are permitted e.g. laptops, phones)

Blank scrap paper permitted - unlimited sheets permitted

Materials To Be Supplied To Students:

None

Instructions To Students:

Additional exam materials (eg. answer booklets, rough paper) will be provided upon request.

Complete all examination questions in the space provided on this examination paper. If more space is required, use the backs of pages. The final page is a formula sheet, which you may detach. Nothing written on the formula sheet will be marked. Answers will be marked based on correctness, quality of expression, depth of understanding demonstrated, and clarity of explanation. If you have any queries about a particular examination question, or if you make any assumptions about the meaning of an examination question, please state this clearly at the start of your solution to the examination question.

For Examiner Use Only

Question Mark

1	
2	
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12	

Total _____

(out of 65)

Each question is worth the number of marks indicated on the right and the total number of marks available is 65.

1. Let p and q be statement variables. Prove that

$$(p \vee q) \rightarrow (q \wedge \sim p) \equiv \sim p.$$

You may use either truth tables or logical equivalences. If you use logical equivalences, you should name the laws you use at each step.

(4 marks)

2. For each of the following statements, determine whether the statement is true or false. Provide a proof if the statement is true and provide a counterexample if the statement is false.

(6 marks)

(a) For all $b, c, d \in \mathbb{Z}^+$, if $d \nmid (b - 2c)$ then $d \nmid b$ or $d \nmid c$.

(b) For all $x, y \in \mathbb{Z}^+$, if $x \equiv y \pmod{5}$ then $2^x \equiv 2^y \pmod{5}$.

3. Consider the sequence $\{a_n\}_{n \geq 1}$ defined recursively as

$$a_1 = 1, a_2 = 3, a_3 = 4 \text{ and } a_k = a_{k-1} + 2a_{k-3} \text{ for each integer } k \geq 4.$$

Use strong mathematical induction to prove that $a_n < 3^n$ for each integer $n \geq 1$.

(6 marks)

4. (a) Prove that for all sets A, B , and C that are subsets of a universal set,

$$A - (B \cup C) = (A - B) - C.$$

(3 marks)

- (b) Consider the sets $S = \{3, 5\}$ and $T = \{2, 3, 7\}$. Indicate whether each of the following statements is true or false; no explanation is needed.

(2 marks)

- (i) $S \subseteq T$
- (ii) $\emptyset \in \mathcal{P}(S)$
- (iii) $\emptyset \in T$
- (iv) $(3, 3, 3) \in S \times T$

5. Consider the function $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ where $\forall (a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+, f((a, b)) = \text{lcm}(a, b)$.

(a) Is f one-one? Explain your answer.

(2 marks)

(b) Is f onto? Explain your answer.

(2 marks)

6. Recall that $[-1, 3] = \{x \in \mathbb{R} : -1 \leq x \leq 3\}$ and that $(2, 4] = \{x \in \mathbb{R} : 2 < x \leq 4\}$. Define a function and use it to prove that $||[-1, 3]| \leq |(2, 4]|$.

(6 marks)

7. Let S be the set of binary strings of length 6. The weight of a binary string is the number of ones it contains, for example $\text{weight}(110110) = 4$. Define a relation ρ on S by

$$\forall s, t \in S, s \rho t \text{ if and only if } \text{weight}(s) \leq \text{weight}(t).$$

- (a) Prove that ρ is not an equivalence relation. (3 marks)

- (b) Prove that ρ is not a partial order relation. (3 marks)

8. Consider the group $(\mathbb{Z}_4 \times \mathbb{Z}_2, +)$.

(a) Prove that $\{(0, 0), (0, 1), (2, 0), (2, 1)\}$ is a subgroup of $(\mathbb{Z}_4 \times \mathbb{Z}_2, +)$.

(4 marks)

(b) Determine all the elements $(x, y) \in \mathbb{Z}_4 \times \mathbb{Z}_2$ for which $\langle(x, y)\rangle$ is isomorphic to $(\mathbb{Z}_4, +)$ and explain your reasoning.

(3 marks)

9. (a) Determine the multiplicative inverse of 7 in the field $\mathbb{Z}_{11}$.

(2 marks)

- (b) Prove that $\mathbb{Z}_{12}$ with addition and multiplication modulo 12 is not a field.

(2 marks)

- 10.** *For each part (a), (b) and (c) of Question 10, you do not need to compute an integer answer, you may give a mathematical expression that answers the question.*

In Queensland, driver licence identification numbers consist of a string of nine digits chosen from $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ with repetition allowed. For the purposes of this question, assume that no other conditions are placed on the number, so any string of nine digits is possible.

- (a) If a driver licence identification number is chosen at random, what is the probability that it will have no repeated digits?

(2 marks)

- (b) How many driver licence identification numbers contain at least two occurrences of the digit 1?

(2 marks)

- (c) Let A be the set of all driver licence identification numbers containing one even digit and eight odd digits. Let B be the set of all driver licence identification numbers containing one digit that is divisible by 3 and eight digits that are not divisible by 3. Determine $|A \cup B|$.

(4 marks)

- 11.** Determine the coefficient of x^5 in the expansion of $\left(3x^2 - \frac{2}{x}\right)^7$, giving your answer as an integer.
- (3 marks)**

- 12. (a)** Let G be the graph with the following adjacency matrix. Is G Eulerian? Explain your answer.

(2 marks)

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

- (b)** Let G be a simple, connected, Eulerian graph with eight vertices and let e be the number of edges in G . Prove that $8 \leq e \leq 24$.

(4 marks)

END OF EXAMINATION

MATH1061/MATH7861 Examination Formula Page

Logical Equivalences

Given any statement variables p , q and r , a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold.

<i>Commutative laws</i>	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
<i>Associative laws</i>	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
<i>Distributive laws</i>	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
<i>Identity laws</i>	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
<i>Negation laws</i>	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
<i>Double negative laws</i>	$\sim(\sim p) \equiv p$	
<i>Idempotent laws</i>	$p \wedge p \equiv p$	$p \vee p \equiv p$
<i>Universal bound laws</i>	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
<i>De Morgan's laws</i>	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
<i>Absorption laws</i>	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
<i>Negations of $\mathbf{t}$ and $\mathbf{c}$</i>	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$

Set Identities For all sets A , B and C that are subsets of a universal set U :

1. Commutative Laws (a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$
2. Associative Laws (a) $(A \cup B) \cup C = A \cup (B \cup C)$
(b) $(A \cap B) \cap C = A \cap (B \cap C)$
3. Distributive Laws (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. Identity Laws (a) $A \cup \emptyset = A$ (b) $A \cap U = A$
5. Complement Laws (a) $A \cup A^c = U$ (b) $A \cap A^c = \emptyset$
6. Double Complement Law $(A^c)^c = A$
7. Idempotent Laws (a) $A \cup A = A$ (b) $A \cap A = A$
8. Universal Bound Laws (a) $A \cup U = U$ (b) $A \cap \emptyset = \emptyset$
9. De Morgan's Laws (a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = A^c \cup B^c$
10. Absorption Laws (a) $A \cup (A \cap B) = A$ (b) $A \cap (A \cup B) = A$
11. Complement of U and $\emptyset$ (a) $U^c = \emptyset$ (b) $\emptyset^c = U$
12. Set Difference Law $A - B = A \cap B^c$

The Quotient-Remainder Theorem Given any integer n and any positive integer d , there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

Unique Factorisation Theorem Given any integer $n > 1$, there exists a positive integer k , distinct prime numbers $p_1, p_2, \dots, p_k$, and positive integers $e_1, e_2, \dots, e_k$ such that $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, and any other expression for n as a product of prime numbers is identical to this except perhaps for the order in which the factors are written.

Schröder-Bernstein Theorem For all sets A and B , if $|A| \leq |B|$ and $|A| \geq |B|$ then $|A| = |B|$.

Binomial Theorem Given any real numbers a and b , and any nonnegative integer n ,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$